

Lecture 3: Unconstrained Optimization

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1 Unconstrained Optimization (Ch17)

Consider a real-valued function of n variables $F: U \subset \mathbb{R}^n \rightarrow \mathbb{R}$. A point $\mathbf{x}^*$ is a

- ▶ **maximizer** of F on U if $F(\mathbf{x}^*) \geq F(\mathbf{x})$ for all $\mathbf{x} \in U$
- ▶ **strict maximizer** if $F(\mathbf{x}^*) > F(\mathbf{x})$ for all $\mathbf{x} \neq \mathbf{x}^* \in U$
- ▶ **local/relative maximizer** if there is a ball $B_r(\mathbf{x}^*)$ about $\mathbf{x}^*$ such that $F(\mathbf{x}^*) \geq F(\mathbf{x})$ for all $\mathbf{x} \in B_r(\mathbf{x}^*) \cap U$
- ▶ **strict local maximizer** if there is a ball $B_r(\mathbf{x}^*)$ about $\mathbf{x}^*$ such that $F(\mathbf{x}^*) > F(\mathbf{x})$ for all $\mathbf{x} \neq \mathbf{x}^* \in B_r(\mathbf{x}^*) \cap U$

Theorem 17.1

Let $F: U \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ be a C^1 function. If $\mathbf{x}^*$ is a local max or min of F in U and if $\mathbf{x}^*$ is an interior point of U , then

$$\frac{\partial F}{\partial x_i}(\mathbf{x}^*) = 0 \quad \text{for } i = 1, \dots, n.$$

Proof: for each x_i , x_i^* is an interior maximizer of $x_i \mapsto F(x_1^*, \dots, x_{i-1}^*, x_i, x_{i+1}^*, \dots, x_n^*)$. Use Theorem 3.3.

To find the local maxs and mins of $F(x, y) = x^3 - y^3 + 9xy$, we compute the first order partial derivatives and set them equal to zero:

$$\frac{\partial F}{\partial x} = 3x^2 + 9y = 0$$

$$\frac{\partial F}{\partial y} = -3y^2 + 9x = 0$$

Solutions to above system of equations are two points $(0, 0)$ and $(3, -3)$. For now, we can conclude that the only candidates for a max or min of F are these two points. We are unable to say whether either of these two is a max or min.

- ▶ A point x^* is a **critical point** if $DF(x^*) = 0$
- ▶ To determine if a critical point is a max or min, we need to use a condition on the second derivatives of F
- ▶ **Hessian** of F :

$$\begin{pmatrix} \frac{\partial^2 F}{\partial x_1^2} & \cdots & \frac{\partial^2 F}{\partial x_n \partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 F}{\partial x_1 \partial x_n} & \cdots & \frac{\partial^2 F}{\partial x_n^2} \end{pmatrix}$$

Theorem 17.2 (Sufficient Conditions)

Let $F: U \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ be a C^2 function whose domain is an open set U in $\mathbb{R}^n$. Suppose $\mathbf{x}^*$ is a critical point of F :

- (a) if the Hessian $D^2F(\mathbf{x}^*)$ is a negative definite symmetric matrix, then $\mathbf{x}^*$ is a strict local max of F ;
- (b) if the Hessian $D^2F(\mathbf{x}^*)$ is a positive definite symmetric matrix, then $\mathbf{x}^*$ is a strict local min of F ;
- (c) if the Hessian $D^2F(\mathbf{x}^*)$ is indefinite, then $\mathbf{x}^*$ is neither a local max nor a local min of F .

- ▶ Equivalent statements based on analytical characterization of positive definite and negative definite matrices: Theorem 17.3-17.5
- ▶ A critical point of F for which the Hessian $D^2F(\mathbf{x}^*)$ is indefinite is called a **saddle point**
- ▶ A saddle point is a min of F in some directions and a max in other directions
- ▶ Example: $F(x_1, x_2) = x_1^2 - x_2^2$

Theorem 17.6

Let $F: U \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ be a C^2 function. Suppose $\mathbf{x}^*$ is an interior point of U and a local max/min of F . Then

$$DF(\mathbf{x}^*) = \mathbf{0}$$

and $D^2F(\mathbf{x}^*)$ is negative/positive **semidefinite**.

Theorem 17.7

Let $F: U \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ be a C^2 function of n variables. Suppose $\mathbf{x}^*$ is an interior point of U and a local max/min of F .

- (a) if $\mathbf{x}^*$ is a local min of F , then $\frac{\partial F}{\partial x_i}(\mathbf{x}^*) = 0$ for $i = 1, \dots, n$ and all the principal minors of the Hessian $D^2F(\mathbf{x}^*)$ are ≥ 0 .
- (b) if $\mathbf{x}^*$ is a local max of F , then $\frac{\partial F}{\partial x_i}(\mathbf{x}^*) = 0$ for $i = 1, \dots, n$ and all the principal minors of the Hessian $D^2F(\mathbf{x}^*)$ of odd order are ≤ 0 and all the principal minors of the Hessian $D^2F(\mathbf{x}^*)$ of even order are ≥ 0 .

- Previously, we computed the critical points of $F(x, y) = x^3 - y^3 + 9xy$ are $(0, 0)$ and $(3, -3)$, by differentiating the first derivatives, we compute the Hessian of F :

$$\begin{pmatrix} F_{xx} & F_{xy} \\ F_{yx} & F_{yy} \end{pmatrix} = \begin{pmatrix} 6x & 9 \\ 9 & -6y \end{pmatrix}$$

- The first order leading principal minor is $F_{xx} = 6x$ and the second order leading principal minor is $\det D^2 F(\mathbf{x}) = -36xy - 81$.
- At $(0, 0)$, these two minors are 0 and -81 , respectively. Since the second order leading principal minor is negative, $(0, 0)$ is a saddle of F , neither a max nor a min.
- At $(3, -3)$, these two minors are 18 and 243. Since these two numbers are positive, $D^2 F(3, -3)$ is positive definite and $(3, -3)$ is a strict local min of F .
- Notice that $(3, -3)$ is not a global min, because at the point $(0, n)$, $F(0, n) = -n^3$, which goes to $-\infty$ as $n \rightarrow \infty$.

Theorem 17.8

Let $F: U \rightarrow \mathbb{R}^1$ be a C^2 function whose domain is a convex open subset U of $\mathbb{R}^n$.

- (a) The following three conditions are equivalent:
 - (i) F is a concave function on U ; and
 - (ii) $F(y) - F(x) \leq DF(x)(y - x)$ for all $x, y \in U$; and
 - (iii) $D^2F(x)$ is negative semidefinite for all $x \in U$
- (b) The following three conditions are equivalent:
 - (i) F is a convex function on U ; and
 - (ii) $F(y) - F(x) \geq DF(x)(y - x)$ for all $x, y \in U$; and
 - (iii) $D^2F(x)$ is positive semidefinite for all $x \in U$
- (c) If F is concave on U and $DF(\mathbf{x}^*) = \mathbf{0}$ for some $\mathbf{x}^* \in U$, then $\mathbf{x}^*$ is a global max of F on U .
- (d) If F is convex on U and $DF(\mathbf{x}^*) = \mathbf{0}$ for some $\mathbf{x}^* \in U$, then $\mathbf{x}^*$ is a global min of F on U .

- ▶ A monopolist faces two distinct and separated markets (e.g., a domestic market and a foreign market), each with its own demand function
 - ▶ supply: Q_i
 - ▶ inverse demand function: $P_i = G_i(Q_i)$
 - ▶ revenue: $Q_i G_i(Q_i)$
 - ▶ production costs: $C(Q_1 + Q_2)$
- ▶ Profit: $F(Q_1, Q_2) = Q_1 G_1(Q_1) + Q_2 G_2(Q_2) - C(Q_1 + Q_2)$
- ▶ Suppose that the firm will produce a positive amount for each market
- ▶ Problem: compute the maxima of the profit function F in the interior of the positive quadrant

- ▶ Then we have

$$\frac{d(Q_1 G_1(Q_1))}{dQ_1} = \frac{d(Q_2 G_2(Q_2))}{dQ_2} = C'(Q_1 + Q_2)$$

- ▶ The marginal revenue in *each* market = marginal cost of the total output

Economic Applications: Discriminating Monopolist

Example

- ▶ $G(Q_1) = 50 - 5Q_1$
- ▶ $G(Q_2) = 100 - 10Q_2$
- ▶ $C(Q) = 90 + 20Q$
- ▶ In order to maximize profits, how much should the monopolist produce for each market?

Economic Applications: Discriminating Monopolist

Example

- ▶ This discriminating monopolist's profit function is

$$F(Q_1, Q_2) = Q_1(50 - 5Q_1) + Q_2(100 - 10Q_2) - (90 + 20(Q_1 + Q_2))$$

- ▶ The critical point of F satisfy:

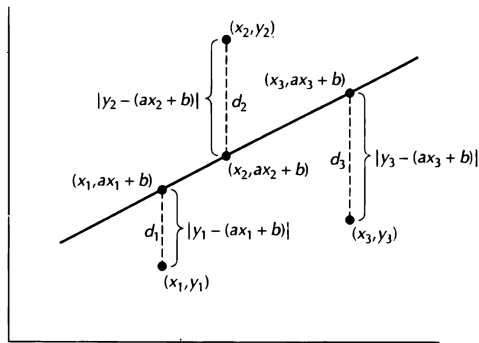
$$\begin{aligned}\frac{\partial F}{\partial Q_1} &= 50 - 10Q_1 - 20 = 0 \\ \frac{\partial F}{\partial Q_2} &= 100 - 20Q_2 - 20 = 0\end{aligned}$$

Hence, $Q_1 = 3$ and $Q_2 = 4$

Economic Applications: Discriminating Monopolist

Example

- ▶ Now check the second order conditions: $F_{Q_1Q_1} = -10$, $F_{Q_2Q_2} = -20$, $F_{Q_1Q_2} = F_{Q_2Q_1} = 0$
- ▶ The first order leading principal minor of $D^2F(3, 4)$ is -10 and second order leading principal minor is 200.
- ▶ Therefore, F is a concave function and the point $(3, 4)$ is a maximizer



The aggregate distance from three data points to a line.

► Objective: $S(m, b) = \sum_{i=1}^n (mx_i + b - y_i)^2$

► FOC:

$$\frac{\partial S}{\partial m} = \sum_{i=1}^n 2(mx_i + b - y_i) \cdot x_i = 0$$

$$\frac{\partial S}{\partial b} = \sum_{i=1}^n 2(mx_i + b - y_i) \cdot 1 = 0$$

► Solving:

$$(\sum_i x_i^2)m + (\sum_i x_i)b = \sum_i x_i y_i$$

$$(\sum_i x_i)m + n \cdot b = \sum_i y_i$$

► Cramer's rule:

$$m^* = \frac{n \sum_i x_i y_i - (\sum_i x_i) (\sum_i y_i)}{n \sum_i x_i^2 - (\sum_i x_i)^2}$$

$$b^* = \frac{(\sum_i x_i^2) (\sum y_i) - (\sum_i x_i) (\sum_i x_i y_i)}{n \sum_i x_i^2 - (\sum_i x_i)^2}$$